



3N+1 PROBLEM 4

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The $3N+1$ sequence is generated using the following rules: If n is even, the next number is $n / 2$. If n is odd, the next number is $3 \times n + 1$. The sequence continues until it reaches 1. The cycle length of a number is the total number of terms in the sequence, including the starting number and 1.

Given two integers i and j , determine the maximum cycle length among all integers between i and j (inclusive).

If $i > j$, swap them before processing.

Input Format

A single line containing two space-separated integers:

i j

Constraints

$1 \leq i, j \leq 1,000,000$

Output Format

Print three integers:

i j maximum_cycle_length

using the original values of i and j .

Sample Input 0

1 10

Sample Output 0

1 10 20

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Language

Java 15



```
3
4 public class Solution {
5     public static int cycleLength(int n){
6         int count=1;
7         while(n!=1){
8             if(n % 2 == 0){
9                 n = n/2;
10            }else{
11                n = 3*n+1;
12            }
13            count++;
14        }
15        return count;
16    }
17    public static void main(String args[]){
```



```
17 // Scanner class for managing input
18 Scanner sc = new Scanner(System.in);
19 int i = sc.nextInt();
20 int j = sc.nextInt();
21 int a = i;
22 int b = j;
23 if(i > j){
24     int temp = i;
25     i = j;
26     j = temp;
27 }
28 int max_cycle = 0;
29 for(int k=i;k<=j;k++){
30     int cycle = cycleLength(k);
31     if(cycle>max_cycle){
32         max_cycle = cycle;
33     }
```

Line: 37 Col: 1



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Test against custom input

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